

Quantitative volcano-monitoring report

# From magma compressibility to surface deformation at La Fossa (Vulcano Island)

Aeolian Islands, southern Tyrrhenian Sea, Italy

Generated on 05 October 2026 by the Volcano Compressibility Agent  
(Freya Mohammadian, 2026)

Method after Freya Mohammadian (2026), *From magma compressibility to surface deformation at La Fossa volcano (Vulcano Island, Italy)*, MSc thesis

**How this report was produced.** The input data were collected automatically from the published literature by an AI research agent (claude-opus-5-5); the physics — EVo volatile equilibria and the Mogi, Yang and Fialko elastic sources — was computed by code that reproduces the La Fossa thesis results to machine precision; the narrative chapters were drafted by the language model from the computed numbers only. Every input carries its source and a status code (M measured, U analytical upper limit, A adopted, D generic default). **0 input(s) are generic defaults or carry caveats** (Section 4.7). Verify key inputs against the cited papers before using the results in any decision.

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## Abstract

This report applies the seven-step compressibility-to-deformation chain to La Fossa (Vulcano Island). The modelled storage levels are Shallow La Fossa reservoir (2.0 km, rhyolite), Intermediate reservoir (5.0 km, trachyte), Deep reservoir (12.0 km, latite). Under the adopted volatile budgets, 1 of the resident magmas are gas-saturated at their storage pressure. [Offline mode: the narrative chapters are written by the language model when an API key is supplied; all tables and figures below are complete.]

# 1 Introduction

## 1.1 Volcano geodesy and the problem of compressible magma

Ground deformation is one of the most direct and quantitative observables of volcanic unrest. InSAR and GNSS now resolve millimetre- to centimetre-scale displacements, and their physical interpretation rests on analytical sources embedded in an elastic half-space: the point-like spherical source of Mogi (1958), the dipping finite prolate spheroid of Yang et al. (1988) as corrected by Newman et al. (2006), and the horizontal penny-shaped crack of Fialko et al. (2001), as implemented in dMODELS (Battaglia et al., 2013).

It is common practice to equate the geodetically inverted cavity-volume change  $\Delta V_c$  with the volume of magma  $V_e$  that entered the reservoir. Mass balance shows that the two are generally different: the pressure rise produced by an injection both compresses the resident magma and dilates the reservoir walls, so only part of the injected volume is expressed as cavity-volume change. For a linearly elastic system,  $V_e = \Delta V_c (1 + \beta_m/\beta_c) = r_V \Delta V_c$  (Rivalta and Segall, 2008), where  $\beta_m$  is the magma compressibility and  $\beta_c$  the chamber compressibility. The chamber term is purely mechanical and shape-dependent; the magma term is controlled above all by exsolved volatiles, because a vapour phase is two to three orders of magnitude more compressible than silicate melt (Kilbride et al., 2016; Segall, 2010).

For monitoring, this physics creates an ambiguity: the spatial wavelength of a deformation field constrains source depth within a specified source geometry and elastic model, but the amplitude is strongly non-unique. One centimetre of uplift may represent a small perturbation of a shallow, gas-rich level or a much larger batch of nearly incompressible magma at depth. Petrological and thermodynamic constraints do not make depth measurable where geodesy cannot; they translate a mechanically inferred cavity-volume change into physically plausible magma-transfer scenarios. This report carries that propagation, from published melt chemistry to predicted surface displacement, for La Fossa (Vulcano Island).

## 1.2 La Fossa (Vulcano Island)

## 1.3 Aim and research questions

## 1.4 Approach

The method is a seven-step chain that translates a storage depth and a magma composition into a predicted surface displacement (Figure 3.1 of the La Fossa thesis; Freya Mohammadian, 2026):

1. From the storage depth  $z$ , the lithostatic pressure  $P = \rho_c gz$ .
2. From  $P$ , temperature and bulk composition, the equilibrium state of the C-O-H-S volatiles — saturation pressure  $P_{\text{sat}}$ , exsolved gas volume fraction  $\phi$  and gas fugacities — computed with the thermodynamic code EVO (Liggins et al., 2020, 2022) using the solubility laws of Burgisser et al. (2015).
3. The magma compressibility, on two limiting branches: the frozen-phase mixture  $\beta_m = \phi\beta_{\text{gas}} + (1 - \phi)\beta_{\text{liquid}}$  (rapid loading) and the equilibrium compressibility  $\beta_m^{\text{eff}} = \Delta \ln \rho_{\text{bulk}} / \Delta P$  along the EVO path (slow loading).
4. The chamber compressibility  $\beta_c$  from the reservoir shape: sphere, prolate spheroid or penny-shaped crack.
5. The volume-partitioning factor  $r_V = 1 + \beta_m/\beta_c$ .
6. The reservoir volume change  $\Delta V_c = V_e/r_V$  for each injected volume.
7. The vertical surface displacement at the centre and at 1 and 2 km with the Mogi, Yang or Fialko source.

The inputs were assembled automatically from the published literature by a research agent; every input carries its source and a status code — measured (M), analytical upper limit (U), adopted representative value (A) or generic default (D) — and generic defaults are listed explicitly in Section 4.7. The chain is then inverted to give, for every configuration, the injected volume and overpressure required for 10 mm of central uplift.

## 2 The volcanic system

### 2.1 Geological setting and eruptive history

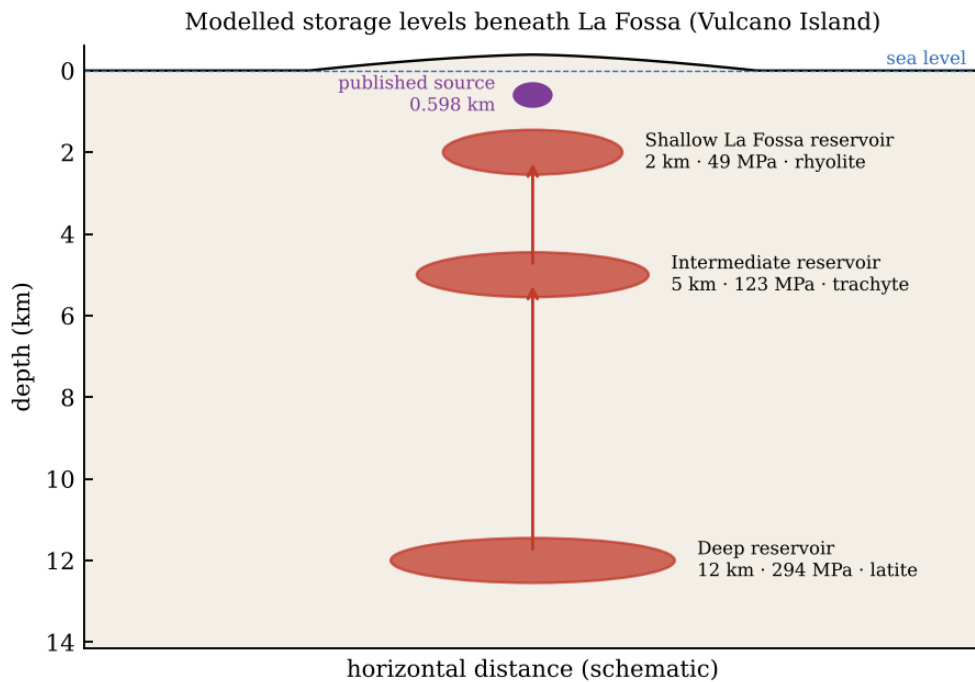
Vulcano is the southernmost island of the Aeolian arc (De Astis et al., 1997).

### 2.2 The magma series

### 2.3 The plumbing system

**Table 2.1.** Storage levels of the plumbing system adopted for the model. Status codes in parentheses (M measured, U upper limit, A adopted, D generic default).

| Level                        | Depth                                      | Published pressure    | Resident / injected magma          | Evidence                                                                                        | Source                    |
|------------------------------|--------------------------------------------|-----------------------|------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------|
| Published deformation source | 0.598 (0.548–0.648) km (M) below sea level | $\Delta P = 8.48$ MPa | Rhyolite (hypothetical) / Trachyte | Expansion of the shallow hydrothermal system fed by magmatic fluids from 3–5 km                 | Di Traglia et al. (2023)  |
| Shallow La Fossa reservoir   | 2 (1.5–2) km (A)                           | 45 (30–60) MPa (M)    | Rhyolite / Trachyte                | Fluid-inclusion barometry on 1888–1890 xenoliths (30–60 MPa); degassing models adopt ~38 MPa    | Clocchiatti et al. (1994) |
| Intermediate reservoir       | 5 (3–5) km (A)                             | 100 MPa (A)           | Trachyte / Latite                  | Upper-crustal density barrier; Vulcanello storage at 3–5 km; latite feeding fumaroles at 3–4 km | Peccerillo et al. (2006)  |
| Deep reservoir               | 12 km (A)                                  | 300 MPa (A)           | Latite / Shoshonite                | Felsic granulite-metapelite boundary; CO <sub>2</sub> -rich fluid inclusions in xenoliths       | Zanon et al. (2003)       |



**Figure 2.1.** Schematic cross-section of the modelled storage levels (depth to scale, horizontal axis schematic), with the lithostatic pressures of Eq. (3.1) and the resident magma of each level.

### 2.4 Volatile budgets and temperatures

**Table 2.2.** Temperatures and volatile budgets with their published ranges and status codes. Values marked U are detection limits assigned as positive numbers; values marked D were not found and are generic defaults.

| Magma      | T                             | H <sub>2</sub> O (wt%)     | CO <sub>2</sub> (wt%) | S (wt%)                    | Sources                                                                  |
|------------|-------------------------------|----------------------------|-----------------------|----------------------------|--------------------------------------------------------------------------|
| Rhyolite   | 1000<br>(1000–1030)<br>°C (M) | 1.25 (1–1.5)<br>(A)        | 0.005 (U)             | 0.02 (U)                   | Clocchiatti et al. (1994); Fusillo et al. (2015); Gioncada et al. (1998) |
| Trachyte   | 1075<br>(1050–1100)<br>°C (A) | 1.25 (1–1.5)<br>(A)        | 0.005 (U)             | 0.07 (M)                   | Clocchiatti et al. (1994); Fusillo et al. (2015); Gioncada et al. (1998) |
| Latite     | 1080<br>(1070–1090)<br>°C (M) | 1.35 (0.8–1.9)<br>(A)      | 0.005 (U)             | 0.1 (M)                    | Gioncada et al. (1998); Paonita et al. (2013); Gioncada et al. (1998)    |
| Shoshonite | 1100 °C (M)                   | 0.85<br>(0.36–1.32)<br>(A) | 0.022 (A)             | 0.03<br>(0.01–0.05)<br>(A) | Fusillo et al. (2015); Paonita et al. (2013)                             |

## 2.5 Hydrothermal system, unrest and ground deformation

Since September 2021 Vulcano has experienced its strongest unrest in decades (Aiuppa et al., 2022; Di Traglia et al., 2023).

## 2.6 Synthesis: the conceptual scheme

### 3 Theoretical framework

Every number of Chapter 5 is produced by the relations below; their derivation and provenance are given in Chapter 3 of the MSc thesis of Freya Mohammadian (2026).

#### 3.1 Pressure and depth

Magma stored at depth  $z$  is assumed to be in lithostatic equilibrium with its overburden:

$$P = \rho_c g z \quad (3.1)$$

with  $\rho_c$  the mean crustal density ( $2500 \text{ kg m}^{-3}$  here). Equation (3.1) fixes the ambient state only; the overpressure  $\Delta P$  that drives deformation is treated separately. For deep reservoirs the reference surface of  $z$  is immaterial, but for a shallow source the depth returned by a geodetic inversion is a coordinate below the model free surface, whereas the pressure depends on the real overburden. Because a gas-bearing magma near its saturation pressure has a gas fraction that varies steeply with  $P$ , the pressure datum of a shallow source is treated here as an explicit modelling assumption and varied (Section 5.6).

#### 3.2 Volatile saturation, speciation and fugacity

Each reservoir is characterised by its total budget of  $\text{H}_2\text{O}$ ,  $\text{CO}_2$  and S. EVo solves the coupled C–O–H–S equilibria between melt and vapour — power-law solubilities calibrated for basaltic, phonolitic and rhyolitic melt classes (Burgisser et al., 2015), homogeneous gas equilibria among  $\text{H}_2\text{O}$ ,  $\text{H}_2$ ,  $\text{CO}_2$ ,  $\text{CO}$ ,  $\text{SO}_2$ ,  $\text{H}_2\text{S}$  and  $\text{S}_2$ , oxygen fugacity buffered relative to FMQ (Frost, 1991) and coupled to the ferric-ferrous equilibrium (Kress and Carmichael, 1991). Run in saturation-finding mode, EVo locates  $P_{\text{sat}}$  for the budget and decompresses at equilibrium to the reservoir pressure. If  $P_{\text{sat}} < P$  the magma is undersaturated and  $\phi = 0$ . The gas volume fraction follows from the gas mass fraction  $w_g$  and the phase densities:

$$\phi = (w_g/\rho_g) / [w_g/\rho_g + (1 - w_g)/\rho_l], \quad \rho_g = P M / (R T) \quad (3.2)$$

with  $M$  the mean molar mass of the equilibrium vapour.

#### 3.3 Magma compressibility

For the vapour, the ideal-gas compressibility is  $\beta_{\text{gas}} = 1/P$ ; for the melt,  $\beta_{\text{liquid}} = (0.5\text{--}2) \times 10^{-10} \text{ Pa}^{-1}$  (Spera, 2000; Rivalta and Segall, 2008), adopted per composition. If a pressure increment is applied faster than volatiles can redistribute, the phase masses are frozen and

$$\beta_m = \phi \beta_{\text{gas}} + (1 - \phi) \beta_{\text{liquid}} \quad (3.3)$$

If loading is slow enough for dissolution equilibrium to be maintained, vapour redissolves as pressure rises and the effective compressibility,  $\beta_m^{\text{eff}} = \Delta \ln \rho_{\text{bulk}} / \Delta P$ , evaluated along the EVo equilibrium path, is always the larger. The two are end-members selected by the ratio of the loading timescale to the diffusive timescale of gas-melt transfer (minutes for water in silicic melt at plausible bubble spacings), so that for unrest developing over weeks to months the equilibrium branch is the more likely. Both are reported wherever a reservoir is gas-bearing; where  $\phi = 0$  they coincide.

#### 3.4 Chamber compressibility

The chamber compressibility  $\beta_c = (1/V_0) dV_c/dP$  depends on shape and is proportional to  $1/\mu$ :

$$\begin{aligned} \text{sphere: } \beta_c &= 3/(4\mu) \\ \text{prolate spheroid: } \beta_c &= (3/4\mu)[A^2/3 - 0.7A + 1.37] \\ \text{penny crack: } \beta_c &= 8(1 - \nu)a^3/(3\mu V_0) \end{aligned} \quad (3.4)$$

The sphere value is exact (Segall, 2010). For the spheroid of aspect ratio  $A = b/a$  the bracket is the shape correction of the dMODELS prolate source (Battaglia et al., 2013), which reproduces the exact Eshelby compliance (Amoruso and Crescentini, 2009) to within about 1.5% across  $0.05 \leq A \leq 1$ . For the penny-shaped crack the Fialko et al. (2001) half-space solution is used, which includes free-surface softening (about 29% at  $a/d = 1$ ). The crack relation requires a thin crack: with mean opening  $\bar{w} = V_0/(\pi a^2)$ , configurations with  $\bar{w}/(2a) > 0.1$  are stiff-walled idealisations rather than sills, and are flagged throughout.

### 3.5 Volume partitioning

An injected volume  $V_e$  is accommodated by cavity dilation and by compression of the resident magma,  $V_e = (\beta_c + \beta_m)V_0\Delta P$ , whence

$$\Delta P = V_e / [(\beta_c + \beta_m)V_0], \quad r_V = 1 + \beta_m/\beta_c, \quad \Delta V_c = V_e/r_V \quad (3.5)$$

For a sphere this reduces to  $r_V = 1 + 4\mu\beta_m/3$  (Rivalta and Segall, 2008). Configurations requiring  $\Delta P$  above 10 MPa for the target signal are described as exceeding a reference screening threshold, a simplified proxy for host-rock strength used only to rank configurations.

### 3.6 Elastic deformation sources

The Mogi point source gives the vertical displacement at radial distance  $r$

$$u_z(r) = (1 - \nu) \Delta V_c d / [\pi (d^2 + r^2)^{3/2}] \quad (3.6)$$

The Yang et al. (1988) spheroid, with the corrections of Newman et al. (2006), and the Fialko et al. (2001) crack are evaluated numerically with dmodels.py, the Python port of dMODELS. All three are linear in source strength, so each source is evaluated once and scaled by  $\Delta V_c$ , and the chain can be inverted by proportionality. All are solutions for a homogeneous, isotropic elastic half-space with a flat surface: topography, layering and time-dependent rheology are not represented.

## 4 Input data, scenario design and verification

### 4.1 Melt compositions

**Table 4.1.** Major-element compositions supplied to EVo (total iron as FeO; Fe<sub>2</sub>O<sub>3</sub> converted with the factor 0.8998). The EVo class is the Burgisser et al. (2015) calibration whose SiO<sub>2</sub> range contains the melt; for intermediate compositions the phonolite class is an approximation.

| Oxide (wt%)                    | Rhyolite                        | Trachyte                        | Latite                          | Shoshonite                      |
|--------------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|
| Sample / unit                  | GS91-50c, 1888-1890 eruption    | Palizzi pumices, TR column      | GS91-17, Pietre Cotte           | GS91-66, Vulcanello I           |
| SiO <sub>2</sub>               | 73.54                           | 61.60                           | 57.45                           | 53.79                           |
| TiO <sub>2</sub>               | 0.13                            | 0.61                            | 0.61                            | 0.70                            |
| Al <sub>2</sub> O <sub>3</sub> | 13.10                           | 17.51                           | 16.83                           | 15.08                           |
| FeO <sub>tot</sub>             | 2.21                            | 4.30                            | 6.60                            | 8.16                            |
| MnO                            | 0.07                            | 0.10                            | 0.14                            | 0.16                            |
| MgO                            | 0.34                            | 1.25                            | 2.58                            | 4.70                            |
| CaO                            | 1.02                            | 2.25                            | 5.17                            | 7.67                            |
| Na <sub>2</sub> O              | 4.35                            | 4.42                            | 3.81                            | 3.55                            |
| K <sub>2</sub> O               | 4.95                            | 7.25                            | 5.67                            | 4.89                            |
| P <sub>2</sub> O <sub>5</sub>  | 0.04                            | 0.22                            | 0.43                            | 0.39                            |
| Status                         | M                               | M                               | M                               | M                               |
| EV0 solubility class           | rhyolite                        | phonolite                       | phonolite                       | basalt                          |
| Source                         | Gioncada et al. (1998), Table 1 | Gioncada et al. (1998), Table 1 | Gioncada et al. (1998), Table 1 | Gioncada et al. (1998), Table 1 |

### 4.2 Magmatic state parameters

**Table 4.2.** Scenario A magmatic state parameters (total budgets supplied to the EVo saturation search; fO<sub>2</sub> buffered at FMQ+1).

| Magma      | T (°C)   | H <sub>2</sub> O (wt%) | CO <sub>2</sub> (wt%) | S (wt%)  | β <sub>liquid</sub> (Pa <sup>-1</sup> ) | Basis                                                                                                                                     |
|------------|----------|------------------------|-----------------------|----------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Rhyolite   | 1000 (M) | 1.25 (A)               | 0.005 (U)             | 0.02 (U) | 1.2×10 <sup>-10</sup>                   | midpoint of 1-1.5 wt% reported for the 1888-1890 magmas; no rhyolite-specific MI data; highest Vulcano detection limit (~50 ppm) assigned |
| Trachyte   | 1075 (A) | 1.25 (A)               | 0.005 (U)             | 0.07 (M) | 1.1×10 <sup>-10</sup>                   | ~1 wt% measured in 1888-1890 trachyte MIs                                                                                                 |
| Latite     | 1080 (M) | 1.35 (A)               | 0.005 (U)             | 0.1 (M)  | 1.0×10 <sup>-10</sup>                   | midpoint of measured 0.8-1.9 wt%; below detection                                                                                         |
| Shoshonite | 1100 (M) | 0.85 (A)               | 0.022 (A)             | 0.03 (A) | 8.0×10 <sup>-11</sup>                   | 220 ppm degassing-model estimate at 100 MPa                                                                                               |

### 4.3 Mechanical parameters and the published source

**Table 4.3.** Mechanical and geometric inputs.

| Parameter                                 | Value                                                                                | Source / note                                         |
|-------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------|
| Crustal density ρ <sub>c</sub>            | 2.5e+03 kg m <sup>-3</sup> (A)                                                       | mean density of edifice and basement; range 2300-2700 |
| Shear modulus, deep μ                     | 10 GPa (A)                                                                           | Heap et al. (2020)                                    |
| Poisson ratio, deep ν                     | 0.25 (A)                                                                             |                                                       |
| Shear modulus, shallow μ                  | 1 GPa (A)                                                                            | Heap et al. (2020); Di Traglia et al. (2023)          |
| Poisson ratio, shallow ν                  | 0.35 (A)                                                                             | Di Traglia et al. (2023)                              |
| Summit elevation                          | 391 m (M)                                                                            | De Astis et al. (1997)                                |
| Caldera half-length (bounds sill radius)  | 2 km (A)                                                                             | Di Traglia et al. (2024)                              |
| Published source: model / depth           | YANG / 0.598 (0.548-0.648) km (M) below sea level                                    | Di Traglia et al. (2023)                              |
| Published source: geometry                | a = 595 m, b = 59 m, plunge 69°, trend 137°                                          | Di Traglia et al. (2023)                              |
| Published source: V <sub>0</sub> , ΔV, ΔP | 8.83×10 <sup>6</sup> m <sup>3</sup> ; 7.31×10 <sup>4</sup> m <sup>3</sup> ; 8.48 MPa | Di Traglia et al. (2023), Table S1                    |

| Parameter                            | Value               | Source / note            |
|--------------------------------------|---------------------|--------------------------|
| Published source: $\mu$ , $\nu$ used | 1 GPa (A), 0.35 (A) | Di Traglia et al. (2023) |

#### 4.4 Scenario design

The scenario matrix crosses the 3 magmatic storage levels with three source geometries — a Mogi sphere and penny-shaped cracks of radius 0.5, 1, 2 km (bounded by the caldera dimension where known) — and with injected volumes of  $10^6$ ,  $5 \times 10^6$  and  $10^7$  m<sup>3</sup> per level. Initial reservoir volumes follow the published values where available and otherwise grow with depth as in the La Fossa scheme ( $5 \times 10^8$ ,  $5 \times 10^9$ ,  $5 \times 10^{10}$  m<sup>3</sup>). The published deformation source is added as the shallowest configuration, with its published geometry and the most evolved magma as a hypothetical magmatic occupant; its pressure is computed at the published model depth and this pressure datum is varied in Section 5.6. The resident composition controls  $\beta_m$ ; the injected composition is a scenario label.

**Table 4.4.** The scenario matrix (Scenario A).

| Configuration                                | $z$ (km) | Resident | Injected   | $V_0$ (m <sup>3</sup> ) | $V_e$ (m <sup>3</sup> )                       | $\frac{\mu}{\nu}$ (GPa), | Thin crack          |
|----------------------------------------------|----------|----------|------------|-------------------------|-----------------------------------------------|--------------------------|---------------------|
| Mogi sphere, 2 km                            | 2        | Rhyolite | Trachyte   | $5.00 \times 10^8$      | $1 \times 10^6, 5 \times 10^6, 1 \times 10^7$ | 10, 0.25                 | n/a                 |
| Mogi sphere, 5 km                            | 5        | Trachyte | Latite     | $5.00 \times 10^9$      | $1 \times 10^6, 5 \times 10^6, 1 \times 10^7$ | 10, 0.25                 | n/a                 |
| Mogi sphere, 12 km                           | 12       | Latite   | Shoshonite | $5.00 \times 10^{10}$   | $1 \times 10^6, 5 \times 10^6, 1 \times 10^7$ | 10, 0.25                 | n/a                 |
| Published source geometry (prolate spheroid) | 0.598    | Rhyolite | Trachyte   | $8.83 \times 10^6$      | $1 \times 10^5, 5 \times 10^5, 1 \times 10^6$ | 1, 0.35                  | n/a                 |
| Penny crack a = 0.5 km, 2 km                 | 2        | Rhyolite | Trachyte   | $5.00 \times 10^8$      | $1 \times 10^6, 5 \times 10^6, 1 \times 10^7$ | 10, 0.25                 | violated            |
| Penny crack a = 0.5 km, 5 km                 | 5        | Trachyte | Latite     | $5.00 \times 10^9$      | $1 \times 10^6, 5 \times 10^6, 1 \times 10^7$ | 10, 0.25                 | violated            |
| Penny crack a = 0.5 km, 12 km                | 12       | Latite   | Shoshonite | $5.00 \times 10^{10}$   | $1 \times 10^6, 5 \times 10^6, 1 \times 10^7$ | 10, 0.25                 | violated            |
| Penny crack a = 1 km, 2 km                   | 2        | Rhyolite | Trachyte   | $5.00 \times 10^8$      | $1 \times 10^6, 5 \times 10^6, 1 \times 10^7$ | 10, 0.25                 | satisfied           |
| Penny crack a = 1 km, 5 km                   | 5        | Trachyte | Latite     | $5.00 \times 10^9$      | $1 \times 10^6, 5 \times 10^6, 1 \times 10^7$ | 10, 0.25                 | violated            |
| Penny crack a = 1 km, 12 km                  | 12       | Latite   | Shoshonite | $5.00 \times 10^{10}$   | $1 \times 10^6, 5 \times 10^6, 1 \times 10^7$ | 10, 0.25                 | violated            |
| Penny crack a = 2 km, 2 km                   | 2        | Rhyolite | Trachyte   | $5.00 \times 10^8$      | $1 \times 10^6, 5 \times 10^6, 1 \times 10^7$ | 10, 0.25                 | satisfied (a/d = 1) |
| Penny crack a = 2 km, 5 km                   | 5        | Trachyte | Latite     | $5.00 \times 10^9$      | $1 \times 10^6, 5 \times 10^6, 1 \times 10^7$ | 10, 0.25                 | satisfied           |
| Penny crack a = 2 km, 12 km                  | 12       | Latite   | Shoshonite | $5.00 \times 10^{10}$   | $1 \times 10^6, 5 \times 10^6, 1 \times 10^7$ | 10, 0.25                 | violated            |

**Table 4.5.** Scenario A (adopted budgets) and the graded volatile-rich Scenario B trials. Scenario B values are trial values, not measurements: they test the dependence of the results on the assumed inventory, with the depth-graded multipliers of the La Fossa thesis.

| Level                      | $z$ (km) | Resident | Trial | H <sub>2</sub> O (wt%) | CO <sub>2</sub> (wt%) | H <sub>2</sub> O / Scenario A |
|----------------------------|----------|----------|-------|------------------------|-----------------------|-------------------------------|
| Shallow La Fossa reservoir | 2        | Rhyolite | A     | 1.25                   | 0.005                 | $\times 1.00$                 |
| Shallow La Fossa reservoir | 2        | Rhyolite | B1    | 2.00                   | 0.010                 | $\times 1.60$                 |
| Shallow La Fossa reservoir | 2        | Rhyolite | B2    | 2.50                   | 0.020                 | $\times 2.00$                 |
| Shallow La Fossa reservoir | 2        | Rhyolite | B3    | 3.00                   | 0.050                 | $\times 2.40$                 |
| Intermediate reservoir     | 5        | Trachyte | A     | 1.25                   | 0.005                 | $\times 1.00$                 |
| Intermediate reservoir     | 5        | Trachyte | B1    | 3.50                   | 0.050                 | $\times 2.80$                 |
| Intermediate reservoir     | 5        | Trachyte | B2    | 4.00                   | 0.100                 | $\times 3.20$                 |
| Intermediate reservoir     | 5        | Trachyte | B3    | 4.50                   | 0.150                 | $\times 3.60$                 |
| Deep reservoir             | 12       | Latite   | A     | 1.35                   | 0.005                 | $\times 1.00$                 |
| Deep reservoir             | 12       | Latite   | B1    | 4.00                   | 0.200                 | $\times 2.96$                 |
| Deep reservoir             | 12       | Latite   | B2    | 5.00                   | 0.300                 | $\times 3.70$                 |

| Level          | z (km) | Resident | Trial | H <sub>2</sub> O (wt%) | CO <sub>2</sub> (wt%) | H <sub>2</sub> O / Scenario A |
|----------------|--------|----------|-------|------------------------|-----------------------|-------------------------------|
| Deep reservoir | 12     | Latite   | B3    | 6.00                   | 0.400                 | ×4.44                         |

## 4.5 Numerical implementation

The chain is implemented in Python. Volatile equilibria are computed with EVO (Liggins et al., 2020, 2022) pinned to the commit used for the La Fossa thesis, in closed-system saturation-finding mode (engine used here: EVO); elastic sources use dmodels.py, the Python port of dMODELS (Battaglia et al., 2013), with the two NumPy-2 compatibility corrections of the thesis. EVO's warning routine for temperatures outside the calibrated range of its H<sub>2</sub>O solubility law is replaced so that EVO continues, as intended, with temperature-independent coefficients; any state affected is listed in Section 4.7. Run on the La Fossa dataset, the same code reproduces every  $r_V$  and uplift value of the thesis to a relative precision of  $10^{-6}$ .

## 4.6 Verification and consistency checks

**Table 4.6.** Verification against closed-form solutions and consistency with the published source mechanics. A consistency check shows that two implementations of the same elastic relation agree; it does not validate the petrological, thermodynamic or elastic assumptions of the chain.

| Test                                                                                            | Kind         | This work | Reference | Agreement                                          |
|-------------------------------------------------------------------------------------------------|--------------|-----------|-----------|----------------------------------------------------|
| Crack $\Delta V$ vs Sneddon full-space solution, $a/d = 0.1$                                    | verification | 1.001     | 1.000     | 0.1%                                               |
| Crack free-surface amplification, $a/d = 1$                                                     | verification | 1.286     | 1.286     | 0.0% (thesis value 1.286)                          |
| Crack central uplift vs point-sill limit ( $a/d = 0.04$ )                                       | verification | 0.9984    | 1.000     | 0.2%                                               |
| Spheroid $\beta_c \cdot \mu$ at $A = 0.99$ vs exact sphere 0.750                                | verification | 0.7528    | 0.7500    | +0.4%                                              |
| Spheroid ( $A = 0.99$ ) central uplift vs Mogi                                                  | verification | 0.9853    | 1.000     | 1.5%                                               |
| Mogi uplift at $r = 0$ vs $(1-\nu)\Delta V/(\pi d^2)$                                           | verification | 1.000     | 1.000     | exact                                              |
| $\beta_c \cdot \mu$ of the source vs value implied by published $\Delta V$ , $\Delta P$ , $V_0$ | consistency  | 0.9780    | 0.9764    | +0.2%                                              |
| $\Delta V_c$ at the published $\Delta P = 8.48$ MPa                                             | consistency  | 73,229    | 73,108    | +0.2%                                              |
| Uplift above the centroid for the published $\Delta V_c$                                        | consistency  | 36.83     | —         | 36.8 mm at centre, 3.8 mm at 1 km, 0.87 mm at 2 km |

## 4.7 Data gaps, defaults and caveats

No input had to be replaced by a generic default.

## 5 Results

### 5.1 Magmatic state of the reservoirs

**Table 5.1.** Magmatic state of each reservoir under Scenario A (EVo).  $\beta_m$  is set by the resident magma; the injected magma is listed for reference.

| Level                      | z (km) | P (MPa) | Role     | Magma      | T (°C) | P <sub>sat</sub> (MPa) | Saturated? | $\phi$ (vol%) | Gas (wt%) |
|----------------------------|--------|---------|----------|------------|--------|------------------------|------------|---------------|-----------|
| Published source level     | 0.598  | 14.67   | resident | Rhyolite   | 1000   | 29.8                   | yes        | 15.54         | 0.216     |
| Published source level     | 0.598  | 14.67   | injected | Trachyte   | 1075   | 53.7                   | yes        | 19.27         | 0.273     |
| Shallow La Fossa reservoir | 2      | 49.05   | resident | Rhyolite   | 1000   | 29.8                   | no         | 0.00          | 0.000     |
| Shallow La Fossa reservoir | 2      | 49.05   | injected | Trachyte   | 1075   | 53.7                   | yes        | 0.01          | 0.001     |
| Intermediate reservoir     | 5      | 122.62  | resident | Trachyte   | 1075   | 53.7                   | no         | 0.00          | 0.000     |
| Intermediate reservoir     | 5      | 122.62  | injected | Latite     | 1080   | 55.9                   | no         | 0.00          | 0.000     |
| Deep reservoir             | 12     | 294.30  | resident | Latite     | 1080   | 55.9                   | no         | 0.00          | 0.000     |
| Deep reservoir             | 12     | 294.30  | injected | Shoshonite | 1100   | 36.0                   | no         | 0.00          | 0.000     |

**Table 5.2.** Equilibrium vapour of the gas-bearing rhyolite at 14.67 MPa (Published source geometry (prolate spheroid));  $\phi = 15.54$  vol%,  $\log fO_2 = FMQ+0.89$ .

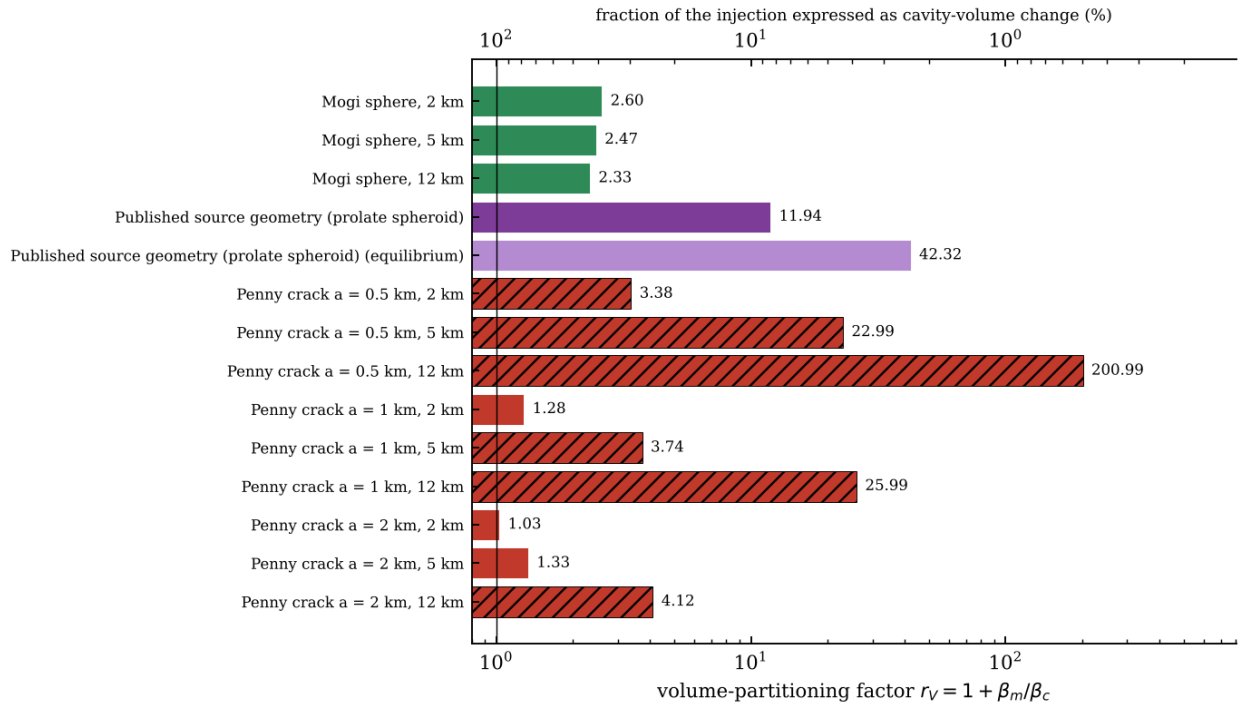
| Species          | mol%                         | Fugacity (bar)                                     |
|------------------|------------------------------|----------------------------------------------------|
| H <sub>2</sub> O | 94.20                        | 136.873                                            |
| H <sub>2</sub> S | 2.13                         | 3.172                                              |
| SO <sub>2</sub>  | 2.09                         | 2.660                                              |
| CO <sub>2</sub>  | 0.99                         | 1.516                                              |
| H <sub>2</sub>   | 0.53                         | —                                                  |
| S <sub>2</sub>   | 0.06                         | 0.088                                              |
| CO               | 0.01                         | —                                                  |
| Bulk             | M = 19.5 g/mol               | $\rho_g = 27.2$ kg/m <sup>3</sup>                  |
| $\beta_m$        | frozen $1.07 \times 10^{-8}$ | equilibrium $4.04 \times 10^{-8}$ Pa <sup>-1</sup> |

### 5.2 Compressibilities and volume partitioning

Across all configurations  $r_V$  ranges from 1.03 to 201 (see Table 5.2 for the geometric status of each).

**Table 5.3.** Compressibilities and volume-partitioning factors. † violates the thin-crack condition (stiff-walled idealisation, not a sill); ‡ crack radius equal to or larger than its depth (validity limit). '=' : the two branches coincide ( $\phi = 0$ ).

| Configuration                                | z (km) | P (MPa) | $\phi$ (vol%) | $\beta_{m1}$ (Pa <sup>-1</sup> ) | $\beta_c$ (Pa <sup>-1</sup> ) | $\beta_c \mu$ | $r_V$ frozen | $r_V$ equil. | 1/ $r_V$ |
|----------------------------------------------|--------|---------|---------------|----------------------------------|-------------------------------|---------------|--------------|--------------|----------|
| Mogi sphere, 2 km                            | 2      | 49.05   | 0.00          | $1.20 \times 10^{-10}$           | $7.50 \times 10^{-11}$        | 0.75          | 2.600        | =            | 38.5%    |
| Mogi sphere, 5 km                            | 5      | 122.62  | 0.00          | $1.10 \times 10^{-10}$           | $7.50 \times 10^{-11}$        | 0.75          | 2.467        | =            | 40.5%    |
| Mogi sphere, 12 km                           | 12     | 294.30  | 0.00          | $1.00 \times 10^{-10}$           | $7.50 \times 10^{-11}$        | 0.75          | 2.333        | =            | 42.9%    |
| Published source geometry (prolate spheroid) | 0.598  | 14.67   | 15.54         | $1.07 \times 10^{-8}$            | $9.78 \times 10^{-10}$        | 0.978         | 11.939       | 42.32        | 8.4%     |
| Penny crack a = 0.5 km, 2 km †               | 2      | 49.05   | 0.00          | $1.20 \times 10^{-10}$           | $5.04 \times 10^{-11}$        | 0.504         | 3.382        | =            | 29.6%    |
| Penny crack a = 0.5 km, 5 km †               | 5      | 122.62  | 0.00          | $1.10 \times 10^{-10}$           | $5.00 \times 10^{-12}$        | 0.05          | 22.988       | =            | 4.4%     |
| Penny crack a = 0.5 km, 12 km †              | 12     | 294.30  | 0.00          | $1.00 \times 10^{-10}$           | $5.00 \times 10^{-13}$        | 0.005         | 200.992      | =            | 0.5%     |
| Penny crack a = 1 km, 2 km                   | 2      | 49.05   | 0.00          | $1.20 \times 10^{-10}$           | $4.21 \times 10^{-10}$        | 4.21          | 1.285        | =            | 77.8%    |
| Penny crack a = 1 km, 5 km †                 | 5      | 122.62  | 0.00          | $1.10 \times 10^{-10}$           | $4.02 \times 10^{-11}$        | 0.402         | 3.739        | =            | 26.7%    |
| Penny crack a = 1 km, 12 km †                | 12     | 294.30  | 0.00          | $1.00 \times 10^{-10}$           | $4.00 \times 10^{-12}$        | 0.04          | 25.992       | =            | 3.8%     |
| Penny crack a = 2 km, 2 km ‡                 | 2      | 49.05   | 0.00          | $1.20 \times 10^{-10}$           | $4.12 \times 10^{-9}$         | 41.2          | 1.029        | =            | 97.2%    |
| Penny crack a = 2 km, 5 km                   | 5      | 122.62  | 0.00          | $1.10 \times 10^{-10}$           | $3.29 \times 10^{-10}$        | 3.29          | 1.334        | =            | 75.0%    |
| Penny crack a = 2 km, 12 km †                | 12     | 294.30  | 0.00          | $1.00 \times 10^{-10}$           | $3.21 \times 10^{-11}$        | 0.321         | 4.118        | =            | 24.3%    |



**Figure 5.1.** Volume-partitioning factor for every configuration (logarithmic scale; upper axis: fraction of an injection expressed as cavity-volume change). Hatched bars violate the thin-crack condition.

### 5.3 Reservoir volume changes

**Table 5.4.** Injected volumes and the resulting reservoir volume changes  $\Delta V_c = V_e/r_v$  (m<sup>3</sup>), frozen branch.

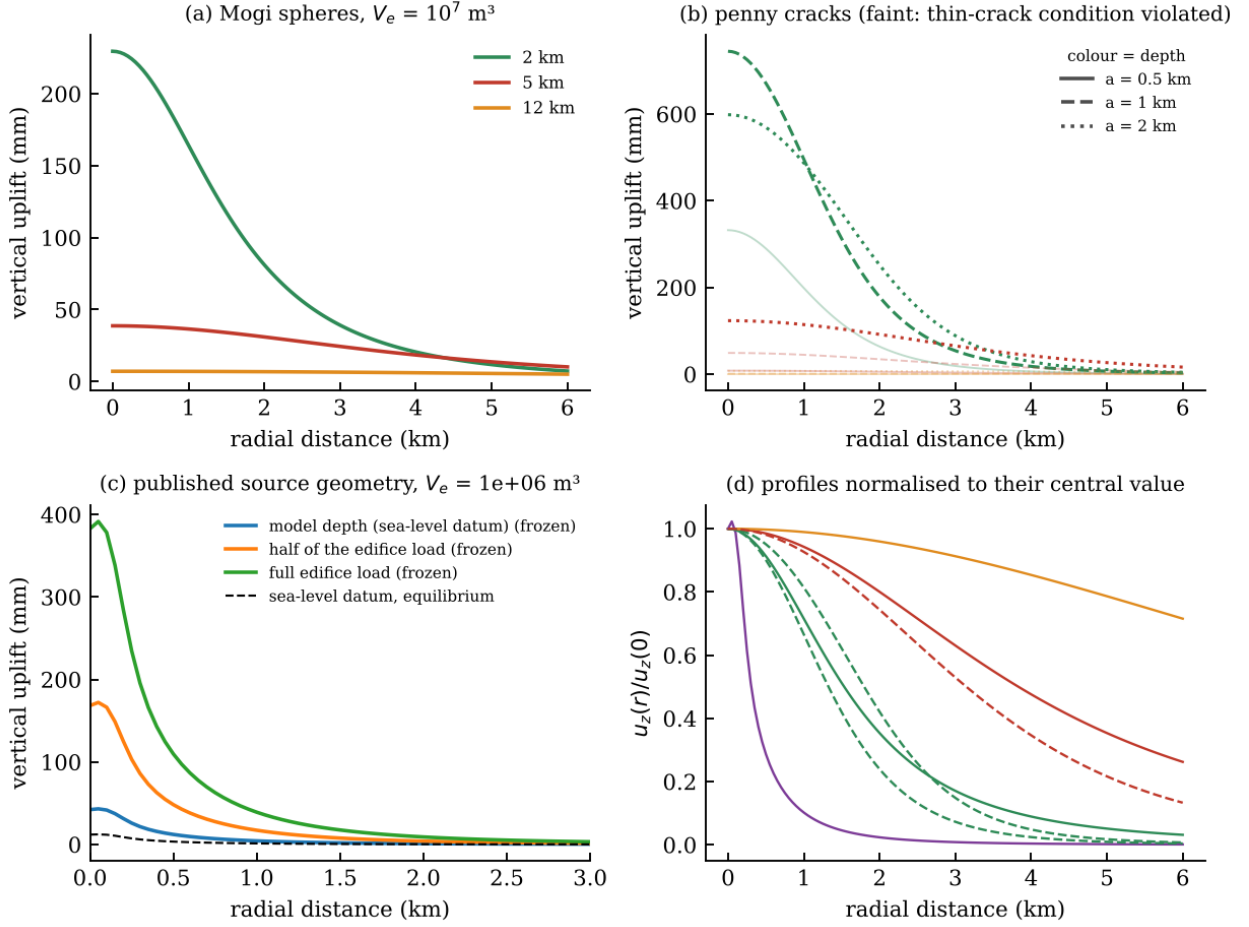
| Configuration                                | $r_v$   | $V_{e1} \rightarrow \Delta V_{c1}$           | $V_{e2} \rightarrow \Delta V_{c2}$           | $V_{e3} \rightarrow \Delta V_{c3}$           | Expressed |
|----------------------------------------------|---------|----------------------------------------------|----------------------------------------------|----------------------------------------------|-----------|
| Mogi sphere, 2 km                            | 2.600   | $1 \times 10^6 \rightarrow 3.85 \times 10^5$ | $5 \times 10^6 \rightarrow 1.92 \times 10^6$ | $1 \times 10^7 \rightarrow 3.85 \times 10^6$ | 38.5%     |
| Mogi sphere, 5 km                            | 2.467   | $1 \times 10^6 \rightarrow 4.05 \times 10^5$ | $5 \times 10^6 \rightarrow 2.03 \times 10^6$ | $1 \times 10^7 \rightarrow 4.05 \times 10^6$ | 40.5%     |
| Mogi sphere, 12 km                           | 2.333   | $1 \times 10^6 \rightarrow 4.29 \times 10^5$ | $5 \times 10^6 \rightarrow 2.14 \times 10^6$ | $1 \times 10^7 \rightarrow 4.29 \times 10^6$ | 42.9%     |
| Published source geometry (prolate spheroid) | 11.939  | $1 \times 10^5 \rightarrow 8.38 \times 10^3$ | $5 \times 10^5 \rightarrow 4.19 \times 10^4$ | $1 \times 10^6 \rightarrow 8.38 \times 10^4$ | 8.4%      |
| Penny crack a = 0.5 km, 2 km †               | 3.382   | $1 \times 10^6 \rightarrow 2.96 \times 10^5$ | $5 \times 10^6 \rightarrow 1.48 \times 10^6$ | $1 \times 10^7 \rightarrow 2.96 \times 10^6$ | 29.6%     |
| Penny crack a = 0.5 km, 5 km †               | 22.988  | $1 \times 10^6 \rightarrow 4.35 \times 10^4$ | $5 \times 10^6 \rightarrow 2.18 \times 10^5$ | $1 \times 10^7 \rightarrow 4.35 \times 10^5$ | 4.4%      |
| Penny crack a = 0.5 km, 12 km †              | 200.992 | $1 \times 10^6 \rightarrow 4.98 \times 10^3$ | $5 \times 10^6 \rightarrow 2.49 \times 10^4$ | $1 \times 10^7 \rightarrow 4.98 \times 10^4$ | 0.5%      |
| Penny crack a = 1 km, 2 km                   | 1.285   | $1 \times 10^6 \rightarrow 7.78 \times 10^5$ | $5 \times 10^6 \rightarrow 3.89 \times 10^6$ | $1 \times 10^7 \rightarrow 7.78 \times 10^6$ | 77.8%     |
| Penny crack a = 1 km, 5 km †                 | 3.739   | $1 \times 10^6 \rightarrow 2.67 \times 10^5$ | $5 \times 10^6 \rightarrow 1.34 \times 10^6$ | $1 \times 10^7 \rightarrow 2.67 \times 10^6$ | 26.7%     |
| Penny crack a = 1 km, 12 km †                | 25.992  | $1 \times 10^6 \rightarrow 3.85 \times 10^4$ | $5 \times 10^6 \rightarrow 1.92 \times 10^5$ | $1 \times 10^7 \rightarrow 3.85 \times 10^5$ | 3.8%      |
| Penny crack a = 2 km, 2 km ‡                 | 1.029   | $1 \times 10^6 \rightarrow 9.72 \times 10^5$ | $5 \times 10^6 \rightarrow 4.86 \times 10^6$ | $1 \times 10^7 \rightarrow 9.72 \times 10^6$ | 97.2%     |
| Penny crack a = 2 km, 5 km                   | 1.334   | $1 \times 10^6 \rightarrow 7.50 \times 10^5$ | $5 \times 10^6 \rightarrow 3.75 \times 10^6$ | $1 \times 10^7 \rightarrow 7.50 \times 10^6$ | 75.0%     |
| Penny crack a = 2 km, 12 km †                | 4.118   | $1 \times 10^6 \rightarrow 2.43 \times 10^5$ | $5 \times 10^6 \rightarrow 1.21 \times 10^6$ | $1 \times 10^7 \rightarrow 2.43 \times 10^6$ | 24.3%     |

### 5.4 Predicted surface displacements

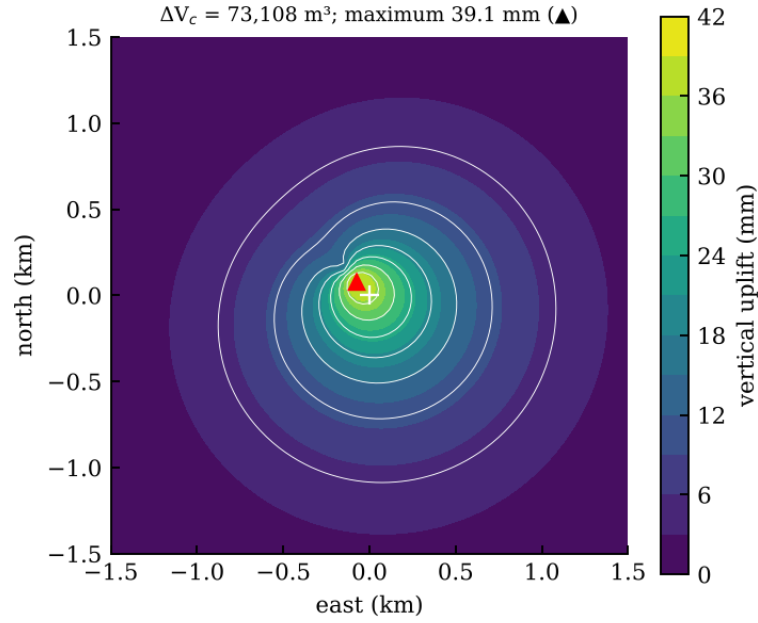
**Table 5.5.** Predicted vertical surface displacement (mm) at the centre and at 1 and 2 km radial distance, for the smallest and largest injected volume (frozen branch; the intermediate volume gives five times the first block). The last column is independent of  $V_e$ ,  $r_v$  and  $\mu$  and indicates depth within a given source model.

| Configuration                                | $z$ (km) | $V_{e\text{small}}$ | 0 km  | 1 km  | 2 km  | $V_{e\text{large}}$ | 0 km  | 1 km  | 2 km  | $u_{2\text{km}}/u_0$ |
|----------------------------------------------|----------|---------------------|-------|-------|-------|---------------------|-------|-------|-------|----------------------|
| Mogi sphere, 2 km                            | 2        | $1 \times 10^6$     | 22.96 | 16.43 | 8.12  | $1 \times 10^7$     | 230   | 164   | 81.16 | 0.35                 |
| Mogi sphere, 5 km                            | 5        | $1 \times 10^6$     | 3.87  | 3.65  | 3.10  | $1 \times 10^7$     | 38.71 | 36.50 | 30.99 | 0.80                 |
| Mogi sphere, 12 km                           | 12       | $1 \times 10^6$     | 0.711 | 0.703 | 0.682 | $1 \times 10^7$     | 7.11  | 7.03  | 6.82  | 0.96                 |
| Published source geometry (prolate spheroid) | 0.598    | $1 \times 10^5$     | 4.22  | 0.431 | 0.100 | $1 \times 10^6$     | 42.20 | 4.31  | 0.999 | 0.02                 |
| Penny crack a = 0.5 km, 2 km †               | 2        | $1 \times 10^6$     | 33.23 | 19.89 | 6.38  | $1 \times 10^7$     | 332   | 199   | 63.84 | 0.19                 |
| Penny crack a = 0.5 km, 5 km †               | 5        | $1 \times 10^6$     | 0.823 | 0.747 | 0.571 | $1 \times 10^7$     | 8.23  | 7.47  | 5.71  | 0.69                 |

| Configuration                     | $z$<br>(km) | $V_e^{\text{small}}$ | 0 km  | 1 km  | 2 km  | $V_e^{\text{large}}$ | 0 km  | 1 km  | 2 km  | $u_{2\text{km}}/u_0$ |
|-----------------------------------|-------------|----------------------|-------|-------|-------|----------------------|-------|-------|-------|----------------------|
| Penny crack $a = 0.5$ km, 12 km † | 12          | $1 \times 10^6$      | 0.016 | 0.016 | 0.015 | $1 \times 10^7$      | 0.165 | 0.162 | 0.154 | 0.93                 |
| Penny crack $a = 1$ km, 2 km      | 2           | $1 \times 10^6$      | 74.48 | 49.59 | 17.83 | $1 \times 10^7$      | 745   | 496   | 178   | 0.24                 |
| Penny crack $a = 1$ km, 5 km †    | 5           | $1 \times 10^6$      | 4.91  | 4.48  | 3.46  | $1 \times 10^7$      | 49.12 | 44.81 | 34.62 | 0.70                 |
| Penny crack $a = 1$ km, 12 km †   | 12          | $1 \times 10^6$      | 0.127 | 0.125 | 0.118 | $1 \times 10^7$      | 1.27  | 1.25  | 1.18  | 0.93                 |
| Penny crack $a = 2$ km, 2 km †    | 2           | $1 \times 10^6$      | 59.85 | 48.72 | 25.19 | $1 \times 10^7$      | 598   | 487   | 252   | 0.42                 |
| Penny crack $a = 2$ km, 5 km      | 5           | $1 \times 10^6$      | 12.35 | 11.45 | 9.19  | $1 \times 10^7$      | 124   | 114   | 91.85 | 0.74                 |
| Penny crack $a = 2$ km, 12 km †   | 12          | $1 \times 10^6$      | 0.784 | 0.771 | 0.734 | $1 \times 10^7$      | 7.84  | 7.71  | 7.34  | 0.94                 |



**Figure 5.2.** Predicted vertical displacement against radial distance for the largest injection of each configuration: (a) Mogi spheres; (b) penny cracks; (c) the published source geometry as a hypothetical magmatic end-member under the pressure datums of Section 5.6; (d) profiles normalised to their central value.

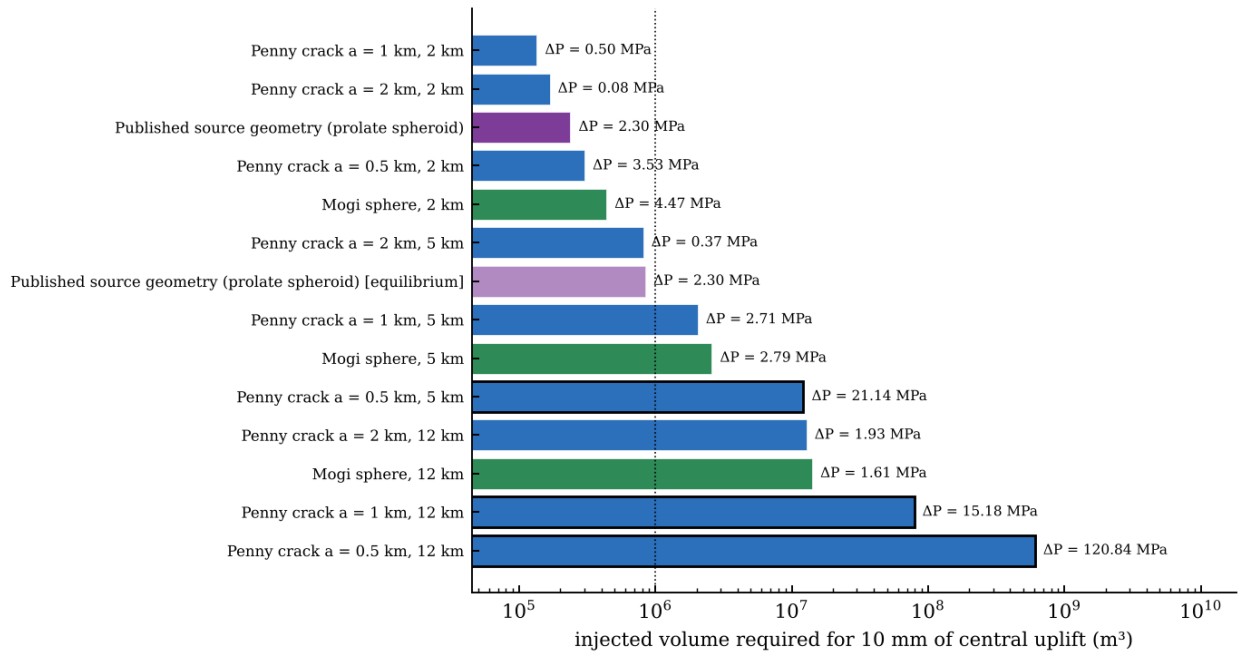


**Figure 5.3.** Map of the vertical displacement produced by the published deformation source driven with its published cavity-volume change (consistency check, not an independent prediction). Cross: epicentre of the centroid.

## 5.5 Detectability: the inverse analysis

**Table 5.6.** Injected volume, cavity-volume change and overpressure required for 10 mm of central uplift, ordered by geodetic cost.  $\Delta V_c$  and  $\Delta P$  are fixed by the elastic problem alone;  $V_e$  depends on the volatile state and the compressibility branch.

| Configuration                                              | $z$<br>(km) | $r_V$  | $V_e$ needed<br>(m <sup>3</sup> ) | $\Delta V_c$ (m <sup>3</sup> ) | $\Delta P$<br>(MPa) | Within 10<br>MPa<br>screen? |
|------------------------------------------------------------|-------------|--------|-----------------------------------|--------------------------------|---------------------|-----------------------------|
| Penny crack $a = 1$ km, 2 km                               | 2           | 1.28   | $1.34 \times 10^5$                | $1.04 \times 10^5$             | 0.50                | yes                         |
| Penny crack $a = 2$ km, 2 km ‡                             | 2           | 1.03   | $1.67 \times 10^5$                | $1.62 \times 10^5$             | 0.08                | yes                         |
| Published source geometry (prolate spheroid)               | 0.598       | 11.94  | $2.37 \times 10^5$                | $1.98 \times 10^4$             | 2.30                | yes                         |
| Penny crack $a = 0.5$ km, 2 km †                           | 2           | 3.38   | $3.01 \times 10^5$                | $8.90 \times 10^4$             | 3.53                | yes                         |
| Mogi sphere, 2 km                                          | 2           | 2.60   | $4.36 \times 10^5$                | $1.68 \times 10^5$             | 4.47                | yes                         |
| Penny crack $a = 2$ km, 5 km                               | 5           | 1.33   | $8.10 \times 10^5$                | $6.07 \times 10^5$             | 0.37                | yes                         |
| Published source geometry (prolate spheroid) [equilibrium] | 0.598       | 42.32  | $8.40 \times 10^5$                | $1.98 \times 10^4$             | 2.30                | yes                         |
| Penny crack $a = 1$ km, 5 km †                             | 5           | 3.74   | $2.04 \times 10^6$                | $5.45 \times 10^5$             | 2.71                | yes                         |
| Mogi sphere, 5 km                                          | 5           | 2.47   | $2.58 \times 10^6$                | $1.05 \times 10^6$             | 2.79                | yes                         |
| Penny crack $a = 0.5$ km, 5 km †                           | 5           | 22.99  | $1.22 \times 10^7$                | $5.29 \times 10^5$             | 21.14               | <b>no</b>                   |
| Penny crack $a = 2$ km, 12 km †                            | 12          | 4.12   | $1.28 \times 10^7$                | $3.10 \times 10^6$             | 1.93                | yes                         |
| Mogi sphere, 12 km                                         | 12          | 2.33   | $1.41 \times 10^7$                | $6.03 \times 10^6$             | 1.61                | yes                         |
| Penny crack $a = 1$ km, 12 km †                            | 12          | 25.99  | $7.89 \times 10^7$                | $3.04 \times 10^6$             | 15.18               | <b>no</b>                   |
| Penny crack $a = 0.5$ km, 12 km †                          | 12          | 200.99 | $6.07 \times 10^8$                | $3.02 \times 10^6$             | 120.84              | <b>no</b>                   |



**Figure 5.4.** Injected volume required for 10 mm of central uplift, with the overpressure each configuration requires. Outlined bars exceed the screening threshold; light bars are the equilibrium branch.

## 5.6 Sensitivity analyses

**Table 5.7.** Volatile-inventory sensitivity: Scenario A against the Scenario B trials, Mogi geometry.

| Level (Mogi)               | Scenario | H <sub>2</sub> O (wt%) | Multiplier | P <sub>sat</sub> (MPa) | φ (vol%) | r <sub>v</sub> | u <sub>z</sub> (0), V <sub>e</sub> = 10 <sup>7</sup> m <sup>3</sup> (mm) |
|----------------------------|----------|------------------------|------------|------------------------|----------|----------------|--------------------------------------------------------------------------|
| Shallow La Fossa reservoir | A        | 1.25                   | ×1.00      | 30                     | 0.00     | 2.60           | 230                                                                      |
| Shallow La Fossa reservoir | B1       | 2.00                   | ×1.60      | 61                     | 0.53     | 4.04           | 148                                                                      |
| Shallow La Fossa reservoir | B2       | 2.50                   | ×2.00      | 96                     | 7.34     | 22.44          | 26.59                                                                    |
| Shallow La Fossa reservoir | B3       | 3.00                   | ×2.40      | 162                    | 17.58    | 50.12          | 11.91                                                                    |
| Intermediate reservoir     | A        | 1.25                   | ×1.00      | 54                     | 0.00     | 2.47           | 38.71                                                                    |
| Intermediate reservoir     | B1       | 3.50                   | ×2.80      | 361                    | 1.37     | 3.94           | 24.26                                                                    |
| Intermediate reservoir     | B2       | 4.00                   | ×3.20      | 524                    | 5.01     | 7.84           | 12.18                                                                    |
| Intermediate reservoir     | B3       | 4.50                   | ×3.60      | 638                    | 9.60     | 12.76          | 7.48                                                                     |
| Deep reservoir             | A        | 1.35                   | ×1.00      | 56                     | 0.00     | 2.33           | 7.11                                                                     |
| Deep reservoir             | B1       | 4.00                   | ×2.96      | 713                    | 0.60     | 2.60           | 6.38                                                                     |
| Deep reservoir             | B2       | 5.00                   | ×3.70      | 852                    | 1.36     | 2.93           | 5.65                                                                     |
| Deep reservoir             | B3       | 6.00                   | ×4.44      | 959                    | 2.91     | 3.61           | 4.59                                                                     |

**Table 5.8.** The published source geometry as a hypothetical magmatic end-member under three pressure datums, for V<sub>e</sub> = 1 × 10<sup>6</sup> m<sup>3</sup>.

| Pressure datum                | Overburden (km) | P (MPa) | φ (vol%) | r <sub>v</sub> frozen | r <sub>v</sub> equil. | u <sub>z</sub> (0) frozen (mm) | u <sub>z</sub> (0) equil. (mm) |
|-------------------------------|-----------------|---------|----------|-----------------------|-----------------------|--------------------------------|--------------------------------|
| model depth (sea-level datum) | 0.598           | 14.67   | 15.54    | 11.94                 | 42.32                 | 42.20                          | 11.90                          |
| half of the edifice load      | 0.793           | 19.46   | 3.57     | 2.99                  | 15.52                 | 168                            | 32.46                          |
| full edifice load             | 0.989           | 24.26   | 0.46     | 1.32                  | 2.93                  | 383                            | 172                            |

**Table 5.9.** Joint sensitivity of the shallow source to the dissolved water content (±20%) and the pressure datum.

| H <sub>2</sub> O (wt%) | Datum                         | P (MPa) | P <sub>sat</sub> (MPa) | φ (vol%) | r <sub>v</sub> frozen | r <sub>v</sub> equil. |
|------------------------|-------------------------------|---------|------------------------|----------|-----------------------|-----------------------|
| 1.00                   | model depth (sea-level datum) | 14.7    | 23.9                   | 2.87     | 3.12                  | 16.25                 |
| 1.00                   | half of the edifice load      | 19.5    | 23.9                   | 0.30     | 1.28                  | 2.34                  |
| 1.00                   | full edifice load             | 24.3    | 23.9                   | 0.00     | 1.12                  | 1.12                  |
| 1.25                   | model depth (sea-level datum) | 14.7    | 29.8                   | 15.54    | 11.94                 | 42.32                 |
| 1.25                   | half of the edifice load      | 19.5    | 29.8                   | 3.57     | 2.99                  | 15.52                 |
| 1.25                   | full edifice load             | 24.3    | 29.8                   | 0.46     | 1.32                  | 2.93                  |
| 1.50                   | model depth (sea-level datum) | 14.7    | 36.5                   | 28.57    | 21.01                 | 49.75                 |

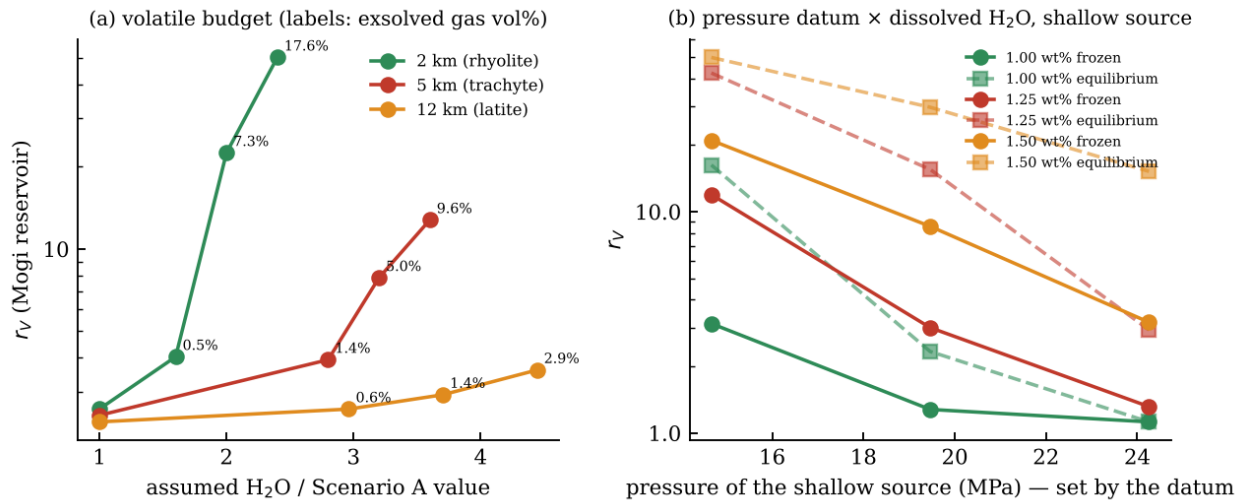
| H <sub>2</sub> O (wt%) | Datum                    | P (MPa) | P <sub>sat</sub> (MPa) | φ (vol%) | r <sub>v</sub> frozen | r <sub>v</sub> equil. |
|------------------------|--------------------------|---------|------------------------|----------|-----------------------|-----------------------|
| 1.50                   | half of the edifice load | 19.5    | 36.5                   | 14.23    | 8.58                  | 29.63                 |
| 1.50                   | full edifice load        | 24.3    | 36.5                   | 4.86     | 3.17                  | 15.21                 |

**Table 5.10.** Other levers: crustal density, redox state, source aspect ratio and elastic parameters.

| Lever value                              | Configuration                     | Effect                                                                | r <sub>v</sub> |
|------------------------------------------|-----------------------------------|-----------------------------------------------------------------------|----------------|
| ρ <sub>c</sub> = 2300 kg m <sup>-3</sup> | published source                  | P = 13.49 MPa, φ = 20.02%                                             | 16.27          |
| ρ <sub>c</sub> = 2500 kg m <sup>-3</sup> | published source                  | P = 14.67 MPa, φ = 15.54%                                             | 11.94          |
| ρ <sub>c</sub> = 2700 kg m <sup>-3</sup> | published source                  | P = 15.84 MPa, φ = 11.92%                                             | 8.80           |
| ΔFMQ = +0.0                              | published source                  | φ = 15.64%, fSO <sub>2</sub> = 0.68 bar, fH <sub>2</sub> S = 5.70 bar | —              |
| ΔFMQ = +1.0                              | published source                  | φ = 15.54%, fSO <sub>2</sub> = 2.66 bar, fH <sub>2</sub> S = 3.17 bar | —              |
| ΔFMQ = +2.0                              | published source                  | φ = 15.49%, fSO <sub>2</sub> = 4.72 bar, fH <sub>2</sub> S = 0.47 bar | —              |
| A = 0.099                                | published source                  | β <sub>c</sub> μ = 0.978                                              | 11.94          |
| A = 0.2                                  | published source                  | β <sub>c</sub> μ = 0.932                                              | 12.47          |
| A = 0.3                                  | published source                  | β <sub>c</sub> μ = 0.892                                              | 12.99          |
| A = 0.5                                  | published source                  | β <sub>c</sub> μ = 0.827                                              | 13.93          |
| A = 0.99                                 | published source                  | β <sub>c</sub> μ = 0.753                                              | 15.21          |
| μ = 5 GPa, ν = 0.25                      | Shallow La Fossa reservoir (Mogi) | u <sub>z</sub> (0) = 332 mm                                           | 1.80           |
| μ = 10 GPa, ν = 0.15                     | Shallow La Fossa reservoir (Mogi) | u <sub>z</sub> (0) = 260 mm                                           | 2.60           |
| μ = 10 GPa, ν = 0.25                     | Shallow La Fossa reservoir (Mogi) | u <sub>z</sub> (0) = 230 mm                                           | 2.60           |
| μ = 10 GPa, ν = 0.35                     | Shallow La Fossa reservoir (Mogi) | u <sub>z</sub> (0) = 199 mm                                           | 2.60           |
| μ = 20 GPa, ν = 0.25                     | Shallow La Fossa reservoir (Mogi) | u <sub>z</sub> (0) = 142 mm                                           | 4.20           |
| μ = 40 GPa, ν = 0.25                     | Shallow La Fossa reservoir (Mogi) | u <sub>z</sub> (0) = 80.65 mm                                         | 7.40           |

**Table 5.11.** Geometric consistency of the penny-shaped configurations (thin-crack condition  $\bar{w}/2a \leq 0.1$ ).

| a (km) | z (km) | V <sub>0</sub> (m <sup>3</sup> ) | $\bar{w}$ (m) | $\bar{w}/2a$ | a/d  | r <sub>v</sub> | Status                 |
|--------|--------|----------------------------------|---------------|--------------|------|----------------|------------------------|
| 0.5    | 2      | 5×10 <sup>8</sup>                | 637           | 0.637        | 0.25 | 3.38           | violated               |
| 0.5    | 5      | 5×10 <sup>9</sup>                | 6,366         | 6.37         | 0.10 | 22.99          | violated               |
| 0.5    | 12     | 5×10 <sup>10</sup>               | 63,662        | 63.7         | 0.04 | 200.99         | violated               |
| 1      | 2      | 5×10 <sup>8</sup>                | 159           | 0.0796       | 0.50 | 1.28           | clean on both criteria |
| 1      | 5      | 5×10 <sup>9</sup>                | 1,592         | 0.796        | 0.20 | 3.74           | violated               |
| 1      | 12     | 5×10 <sup>10</sup>               | 15,915        | 7.96         | 0.08 | 25.99          | violated               |
| 2      | 2      | 5×10 <sup>8</sup>                | 40            | 0.00995      | 1.00 | 1.03           | satisfied; a/d >= 1    |
| 2      | 5      | 5×10 <sup>9</sup>                | 398           | 0.0995       | 0.40 | 1.33           | clean on both criteria |
| 2      | 12     | 5×10 <sup>10</sup>               | 3,979         | 0.995        | 0.17 | 4.12           | violated               |



**Figure 5.5.** Sensitivity of the volume-partitioning factor: (a) to the assumed volatile budget of each Mogi reservoir; (b) to the pressure datum and dissolved water content of the shallow source, on both compressibility branches.

## 6 Discussion

## 7 Conclusions and future work

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## Appendix A Complete numerical results

**Table A.1.** Complete forward results (largest injected volume; displacements in mm; compressibilities in Pa<sup>-1</sup>).

| Configuration                                | P<br>(MPa) | $\varphi$ | $\beta_m$               | $\beta_m^{\text{eff}}$  | $\beta_c$               | $r_v$    | $r_v^{\text{eq}}$ | $\Delta V_{c3}$     | u(0)   | u(1<br>km) | u(2<br>km) |
|----------------------------------------------|------------|-----------|-------------------------|-------------------------|-------------------------|----------|-------------------|---------------------|--------|------------|------------|
| Mogi sphere, 2 km                            | 49.050     | 0.00000   | $1.200 \times 10^{-10}$ | $1.200 \times 10^{-10}$ | $7.500 \times 10^{-11}$ | 2.6000   | 2.6000            | $3.846 \times 10^6$ | 229.6  | 164.3      | 81.16      |
| Mogi sphere, 5 km                            | 122.625    | 0.00000   | $1.100 \times 10^{-10}$ | $1.100 \times 10^{-10}$ | $7.500 \times 10^{-11}$ | 2.4667   | 2.4667            | $4.054 \times 10^6$ | 38.71  | 36.5       | 30.99      |
| Mogi sphere, 12 km                           | 294.300    | 0.00000   | $1.000 \times 10^{-10}$ | $1.000 \times 10^{-10}$ | $7.500 \times 10^{-11}$ | 2.3333   | 2.3333            | $4.286 \times 10^6$ | 7.105  | 7.032      | 6.819      |
| Published source geometry (prolate spheroid) | 14.666     | 0.15541   | $1.070 \times 10^{-8}$  | $4.041 \times 10^{-8}$  | $9.780 \times 10^{-10}$ | 11.9392  | 42.3242           | $8.376 \times 10^4$ | 42.2   | 4.308      | 0.9987     |
| Penny crack a = 0.5 km, 2 km                 | 49.050     | 0.00000   | $1.200 \times 10^{-10}$ | $1.200 \times 10^{-10}$ | $5.039 \times 10^{-11}$ | 3.3816   | 3.3816            | $2.957 \times 10^6$ | 332.3  | 198.9      | 63.84      |
| Penny crack a = 0.5 km, 5 km                 | 122.625    | 0.00000   | $1.100 \times 10^{-10}$ | $1.100 \times 10^{-10}$ | $5.003 \times 10^{-12}$ | 22.9885  | 22.9885           | $4.350 \times 10^5$ | 8.226  | 7.47       | 5.707      |
| Penny crack a = 0.5 km, 12 km                | 294.300    | 0.00000   | $1.000 \times 10^{-10}$ | $1.000 \times 10^{-10}$ | $5.000 \times 10^{-13}$ | 200.9923 | 200.9923          | $4.975 \times 10^4$ | 0.1647 | 0.1619     | 0.1538     |
| Penny crack a = 1 km, 2 km                   | 49.050     | 0.00000   | $1.200 \times 10^{-10}$ | $1.200 \times 10^{-10}$ | $4.211 \times 10^{-10}$ | 1.2850   | 1.2850            | $7.782 \times 10^6$ | 744.8  | 495.9      | 178.3      |
| Penny crack a = 1 km, 5 km                   | 122.625    | 0.00000   | $1.100 \times 10^{-10}$ | $1.100 \times 10^{-10}$ | $4.016 \times 10^{-11}$ | 3.7389   | 3.7389            | $2.675 \times 10^6$ | 49.12  | 44.81      | 34.62      |
| Penny crack a = 1 km, 12 km                  | 294.300    | 0.00000   | $1.000 \times 10^{-10}$ | $1.000 \times 10^{-10}$ | $4.001 \times 10^{-12}$ | 25.9924  | 25.9924           | $3.847 \times 10^5$ | 1.267  | 1.245      | 1.184      |
| Penny crack a = 2 km, 2 km                   | 49.050     | 0.00000   | $1.200 \times 10^{-10}$ | $1.200 \times 10^{-10}$ | $4.116 \times 10^{-9}$  | 1.0292   | 1.0292            | $9.717 \times 10^6$ | 598.5  | 487.2      | 251.9      |
| Penny crack a = 2 km, 5 km                   | 122.625    | 0.00000   | $1.100 \times 10^{-10}$ | $1.100 \times 10^{-10}$ | $3.293 \times 10^{-10}$ | 1.3341   | 1.3341            | $7.496 \times 10^6$ | 123.5  | 114.5      | 91.85      |
| Penny crack a = 2 km, 12 km                  | 294.300    | 0.00000   | $1.000 \times 10^{-10}$ | $1.000 \times 10^{-10}$ | $3.208 \times 10^{-11}$ | 4.1176   | 4.1176            | $2.429 \times 10^6$ | 7.835  | 7.707      | 7.34       |

## Appendix B Provenance of every input

**Table B.1.** Provenance and status of every input. M = measured; U = analytical upper limit; A = adopted representative value; D = generic default (not volcano-specific).

| Input                            | Value                                  | Status | Source                                        | Note                                                                                  |
|----------------------------------|----------------------------------------|--------|-----------------------------------------------|---------------------------------------------------------------------------------------|
| Rhyolite composition             | GS91-50c, 1888-1890 eruption           | M      | Gioncada et al. (1998), Table 1               |                                                                                       |
| Rhyolite T                       | 1000 °C                                | M      | Clocchiatti et al. (1994)                     |                                                                                       |
| Rhyolite H <sub>2</sub> O        | 1.25 wt%                               | A      | Clocchiatti et al. (1994)                     | midpoint of 1-1.5 wt% reported for the 1888-1890 magmas; no rhyolite-specific MI data |
| Rhyolite CO <sub>2</sub>         | 0.005 wt%                              | U      | Fusillo et al. (2015)                         | highest Vulcano detection limit (~50 ppm) assigned                                    |
| Rhyolite S                       | 0.02 wt%                               | U      | Gioncada et al. (1998)                        | below detection in rhyolitic groundmass; assigned                                     |
| Rhyolite $\beta_{\text{liquid}}$ | $1.2 \times 10^{-10}$ Pa <sup>-1</sup> | A      | Spera (2000); Rivalta and Segall (2008)       |                                                                                       |
| Trachyte composition             | Palizzi pumices, TR column             | M      | Gioncada et al. (1998), Table 1               |                                                                                       |
| Trachyte T                       | 1075 °C                                | A      | Clocchiatti et al. (1994)                     |                                                                                       |
| Trachyte H <sub>2</sub> O        | 1.25 wt%                               | A      | Clocchiatti et al. (1994)                     | ~1 wt% measured in 1888-1890 trachyte MIs                                             |
| Trachyte CO <sub>2</sub>         | 0.005 wt%                              | U      | Fusillo et al. (2015)                         |                                                                                       |
| Trachyte S                       | 0.07 wt%                               | M      | Gioncada et al. (1998)                        |                                                                                       |
| Trachyte $\beta_{\text{liquid}}$ | $1.1 \times 10^{-10}$ Pa <sup>-1</sup> | A      | Spera (2000); Rivalta and Segall (2008)       |                                                                                       |
| Latite composition               | GS91-17, Pietre Cotte                  | M      | Gioncada et al. (1998), Table 1               |                                                                                       |
| Latite T                         | 1080 °C                                | M      | Gioncada et al. (1998); Paonita et al. (2013) |                                                                                       |
| Latite H <sub>2</sub> O          | 1.35 wt%                               | A      | Gioncada et al. (1998)                        | midpoint of measured 0.8-1.9 wt%                                                      |
| Latite CO <sub>2</sub>           | 0.005 wt%                              | U      | Gioncada et al. (1998)                        | below detection                                                                       |
| Latite S                         | 0.1 wt%                                | M      | Gioncada et al. (1998)                        |                                                                                       |
| Latite $\beta_{\text{liquid}}$   | $1 \times 10^{-10}$ Pa <sup>-1</sup>   | A      | Spera (2000); Rivalta and Segall (2008)       |                                                                                       |
| Shoshonite composition           | GS91-66, Vulcanello I                  | M      | Gioncada et al. (1998), Table 1               |                                                                                       |
| Shoshonite T                     | 1100 °C                                | M      | Fusillo et al. (2015)                         |                                                                                       |

| Input                                     | Value                  | Status | Source                                       | Note                                                  |
|-------------------------------------------|------------------------|--------|----------------------------------------------|-------------------------------------------------------|
| Shoshonite H <sub>2</sub> O               | 0.85 wt%               | A      | Fusillo et al. (2015)                        |                                                       |
| Shoshonite CO <sub>2</sub>                | 0.022 wt%              | A      | Paonita et al. (2013)                        | 220 ppm degassing-model estimate at 100 MPa           |
| Shoshonite S                              | 0.03 wt%               | A      | Fusillo et al. (2015)                        |                                                       |
| Shoshonite $\beta$ liquid                 | 8e-11 Pa <sup>-1</sup> | A      | Spera (2000); Rivalta and Segall (2008)      |                                                       |
| Shallow La Fossa reservoir depth          | 2 km                   | A      | Clocchiatti et al. (1994)                    |                                                       |
| Shallow La Fossa reservoir V <sub>0</sub> | 5.00×10 <sup>8</sup>   | A      |                                              | reference scheme                                      |
| Intermediate reservoir depth              | 5 km                   | A      | Peccerillo et al. (2006)                     |                                                       |
| Intermediate reservoir V <sub>0</sub>     | 5.00×10 <sup>9</sup>   | A      |                                              | reference scheme                                      |
| Deep reservoir depth                      | 12 km                  | A      | Zanon et al. (2003)                          |                                                       |
| Deep reservoir V <sub>0</sub>             | 5.00×10 <sup>10</sup>  | A      |                                              | reference scheme                                      |
| $\rho_{\text{crust}}$                     | 2500                   | A      |                                              | mean density of edifice and basement; range 2300-2700 |
| $\mu$ deep (GPa)                          | 10                     | A      | Heap et al. (2020)                           | intact-rock end of 1-10 GPa                           |
| $\nu$ deep                                | 0.25                   | A      |                                              |                                                       |
| $\mu$ shallow (GPa)                       | 1                      | A      | Heap et al. (2020); Di Traglia et al. (2023) | damaged, hydrothermally altered rock                  |
| $\nu$ shallow                             | 0.35                   | A      | Di Traglia et al. (2023)                     |                                                       |
| $\Delta\text{FMQ}$                        | 1                      | A      |                                              | oxidised arc magmas                                   |

## Appendix C Research log and number audit

Research model: claude-opus-5-5. Usage: offline. Report generated 2026-10-05 16:26.

**Number audit.** Every number in the model-written chapters was compared with the computed facts and the inputs (allowing for the rounding implied by the digits written and for % and km/m conversions). All numbers were matched.

**Web sources returned during the research** (not all were used; the cited references are listed above):